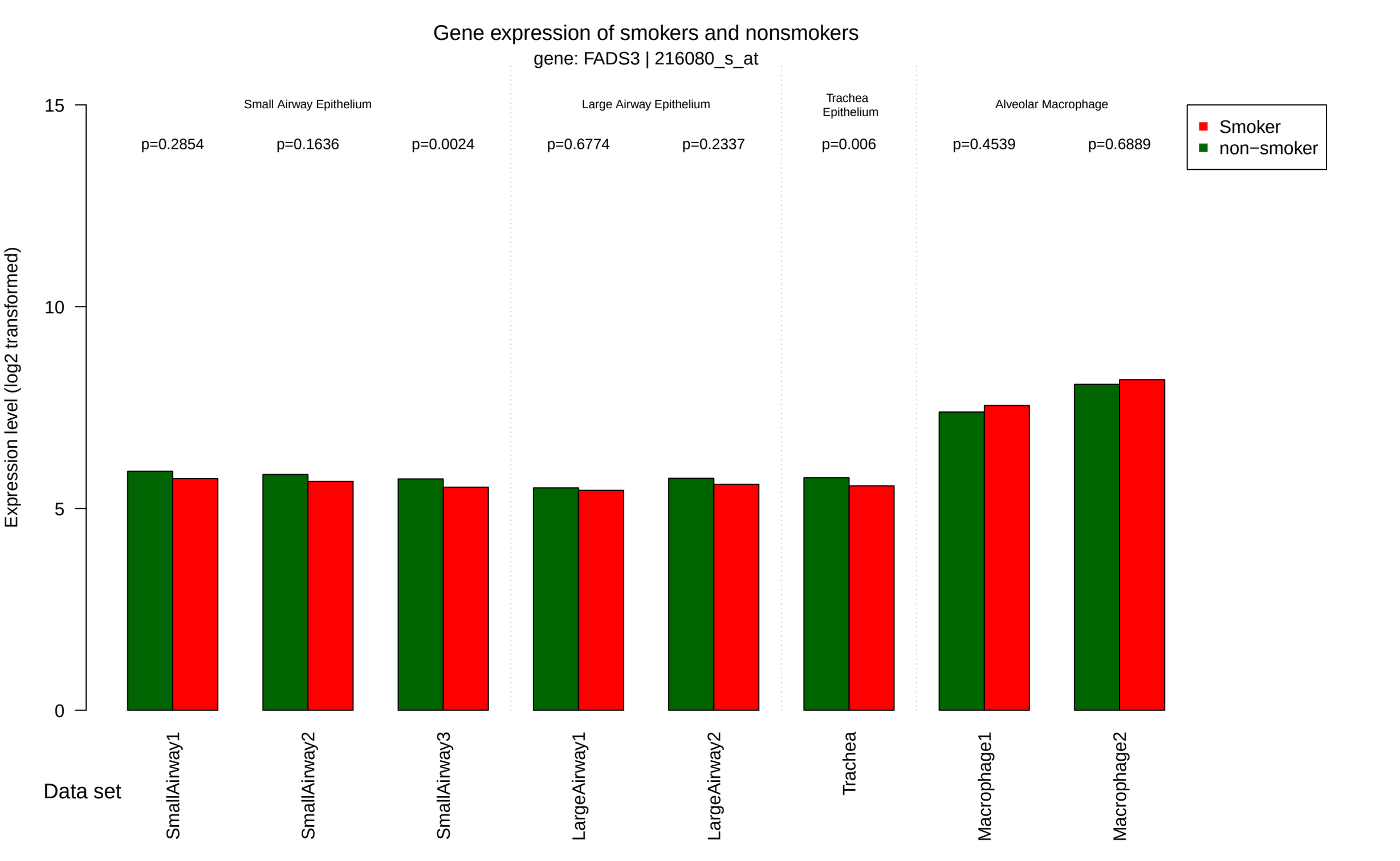
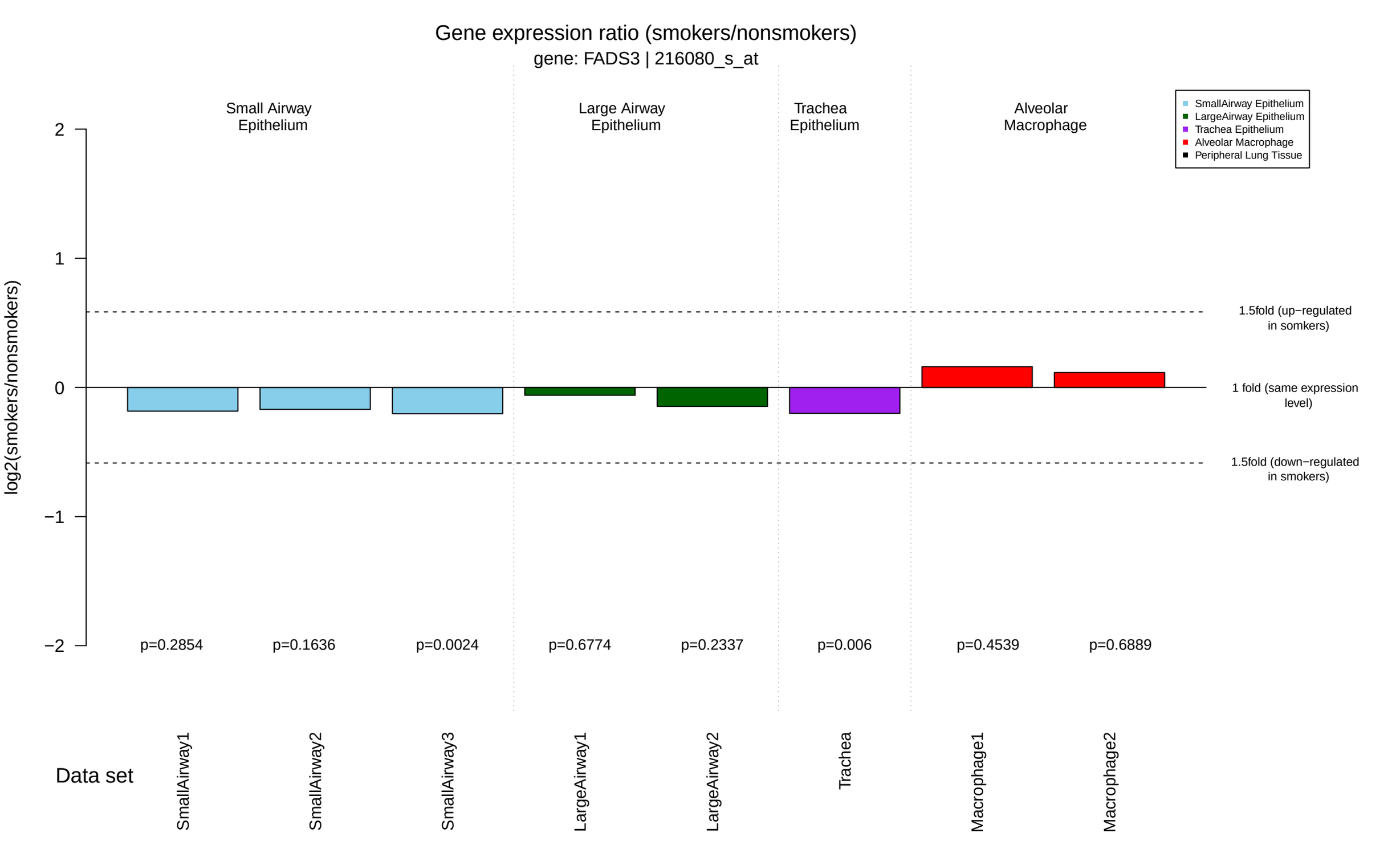
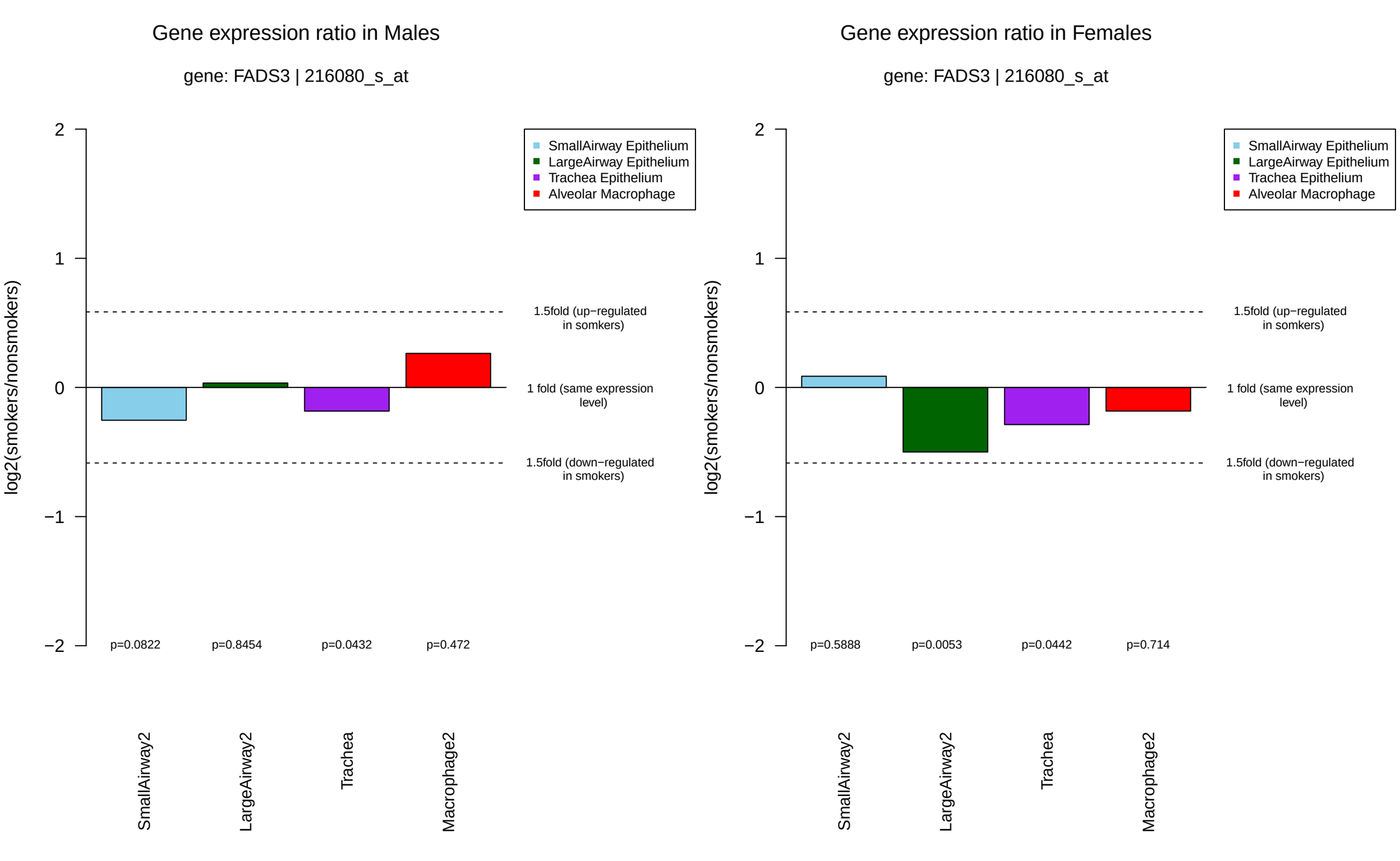
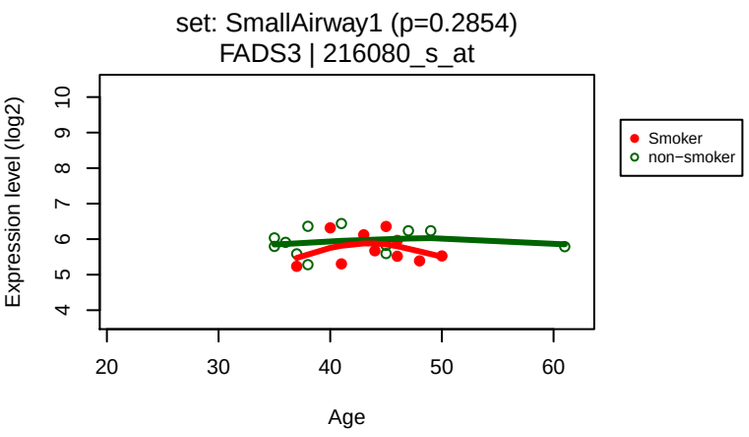
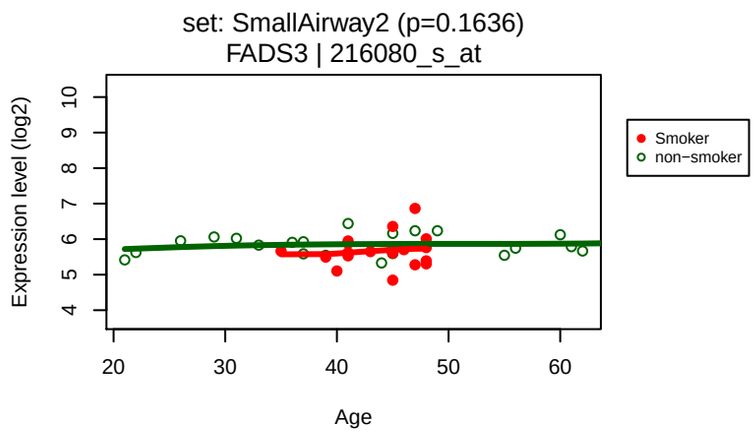
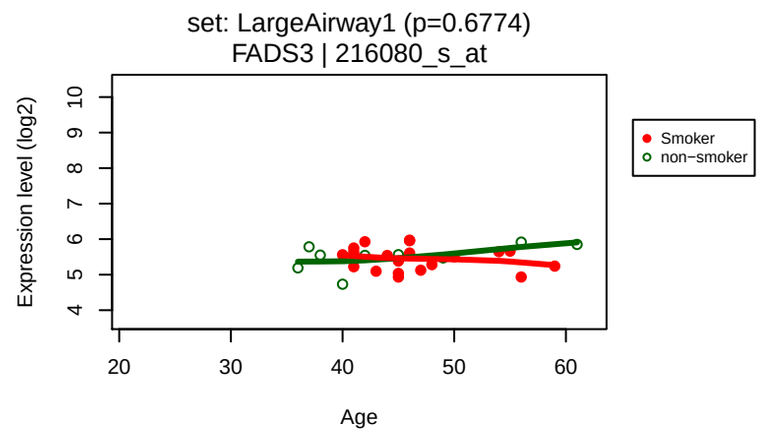

Gene expression of smokers and nonsmokers
gene: FADS3 | 216080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2854
p=0.1636
p=0.0024
p=0.6774
p=0.2337
p=0.006
p=0.4539
p=0.6889
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

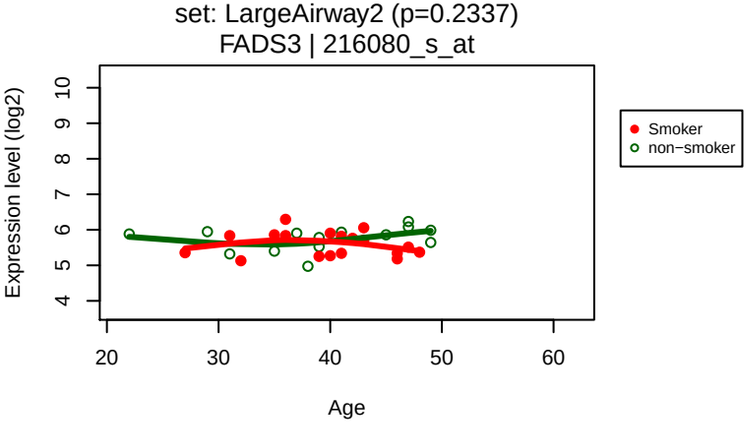

Gene expression ratio (smokers/nonsmokers)
gene: FADS3 | 216080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2854
p=0.1636
p=0.0024
p=0.6774
p=0.2337
p=0.006
p=0.4539
p=0.6889
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

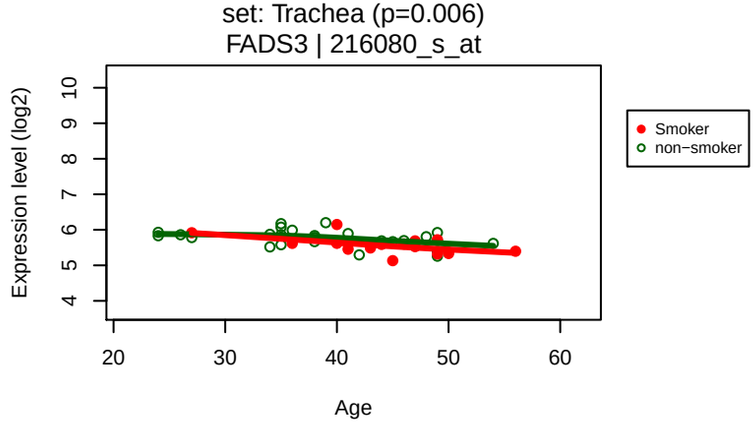

Gene expression ratio in Males
gene: FADS3 | 216080_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0822
p=0.8454
p=0.0432
p=0.472
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FADS3 | 216080_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5888
p=0.0053
p=0.0442
p=0.714
SmallAirway2
LargeAirway2
Trachea
Macrophage2

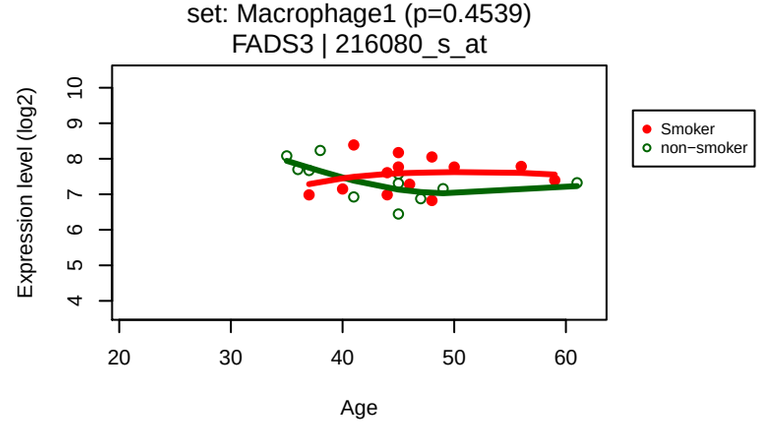

set: SmallAirway1 (p=0.2854)
FADS3 | 216080_s_at
Expression level (log2)
Age
Smoker
non−smoker

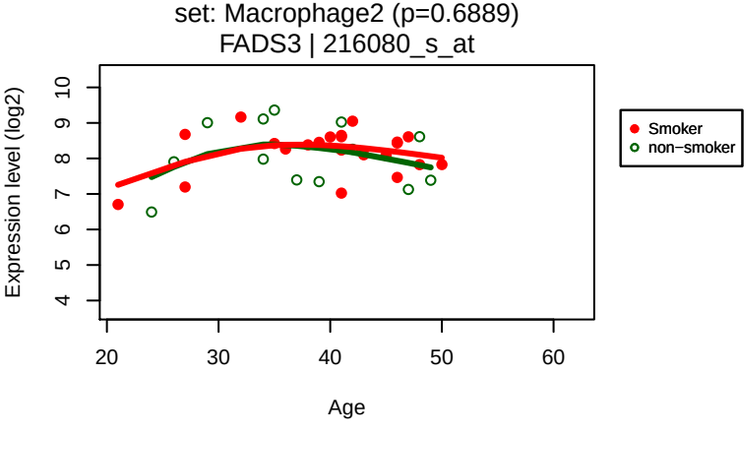

set: SmallAirway2 (p=0.1636)
FADS3 | 216080_s_at
Expression level (log2)
Age
Smoker
non−smoker

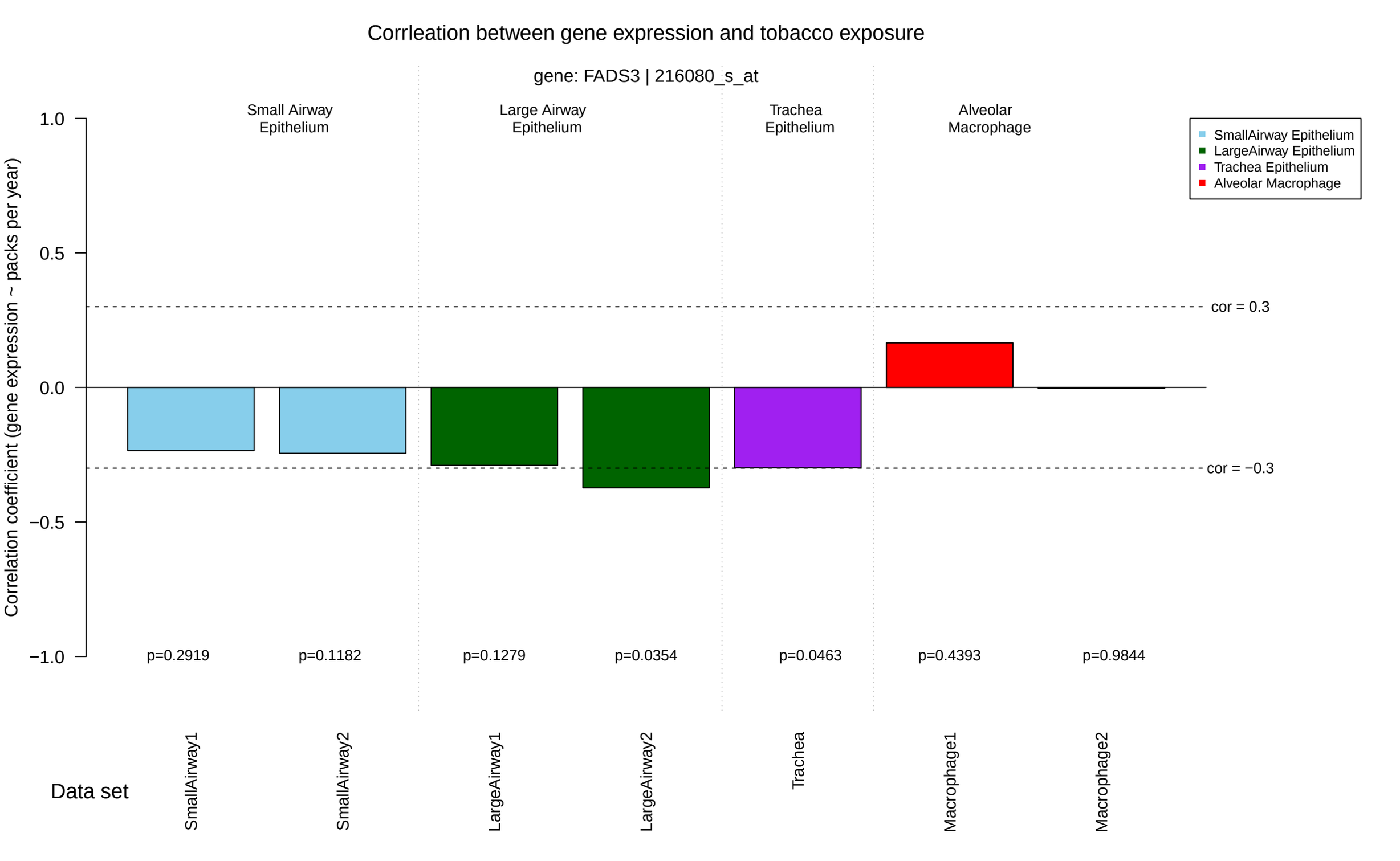

set: LargeAirway1 (p=0.6774)
FADS3 | 216080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2337)
FADS3 | 216080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.006)
FADS3 | 216080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4539)
FADS3 | 216080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6889)
FADS3 | 216080_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FADS3 | 216080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2919
p=0.1182
p=0.1279
p=0.0354
p=0.0463
p=0.4393
p=0.9844
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2